

3. Random Variables & Distributions

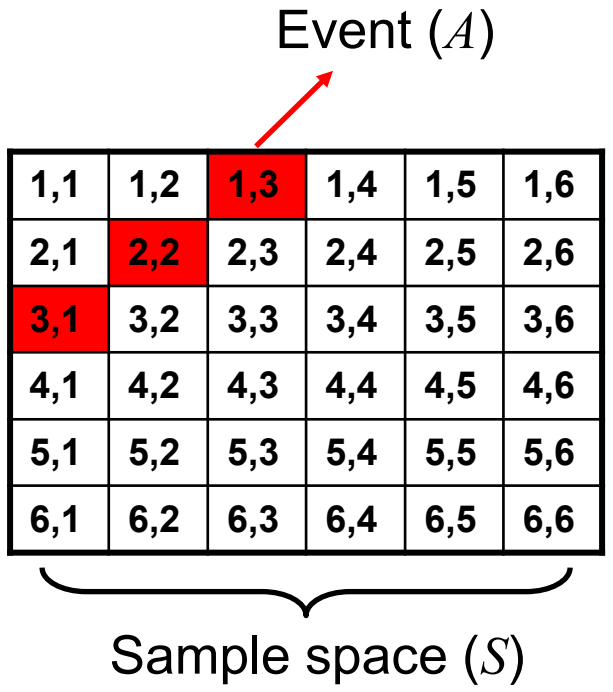
Learning Outcomes

- By the end of this week, you should be able to:

- Understand why and how to use *Random Variables* (*RV*)
- Understand the concept of a *Distribution* of an RV
- Calculate probabilities of discrete RVs using
  - *Probability Mass Functions* (*PMF*)
  - *Cumulative Distribution Functions* (*CDF*)
- Employ common discrete distributions
  - *Bernoulli, Binomial, Hypergeometric*

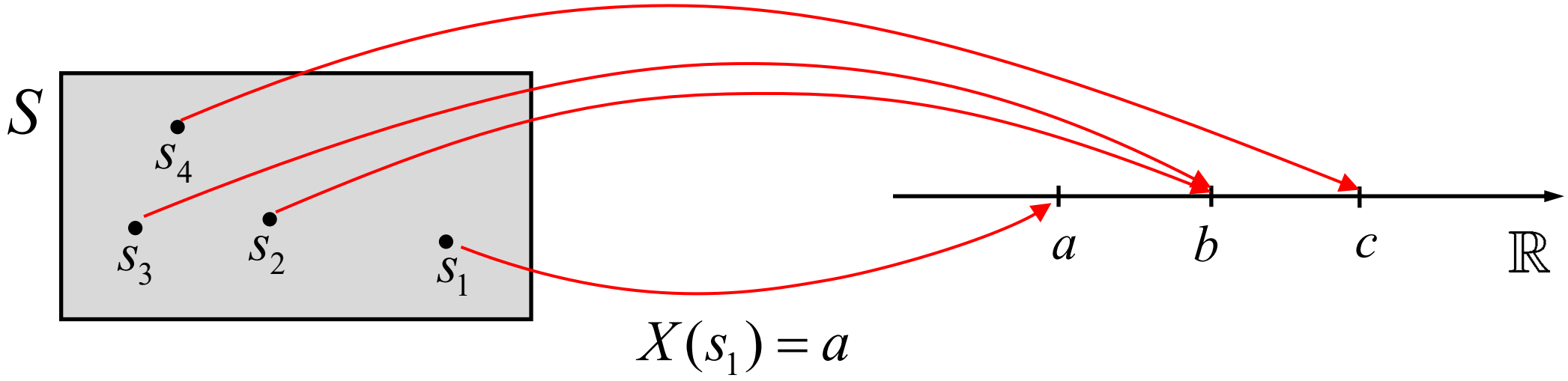
Random Variables

- Consider experiment of rolling 2 dice
  - Sample space $|S| = 6^2 = 36$
- Assume you are only interested in the *sum* of the two rolls
  - E.g., want probability of event $A = \{$"Sum of two rolls is 4"$\}$
- Alternatively, we can use a *Random Variable* (RV) to describe events *and* calculate their probabilities



RV Definition

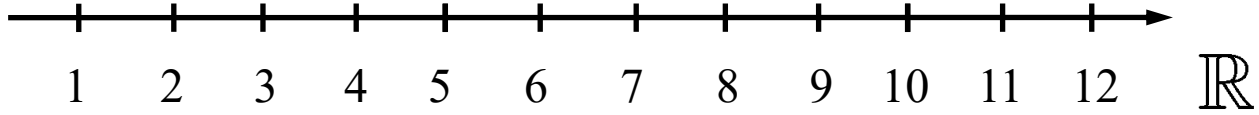
- A *Random Variable* is a *mapping/function* from the sample space (S) to the real line ($\mathbb{R}$)
  - Essentially an RV *assigns numbers to outcomes*
  - RVs are denoted by capital letters (usually X, Y, Z), and their values by lowercase letters



Example

- Let RV X be *sum* of two dice
  - Map outcomes to reals according to X

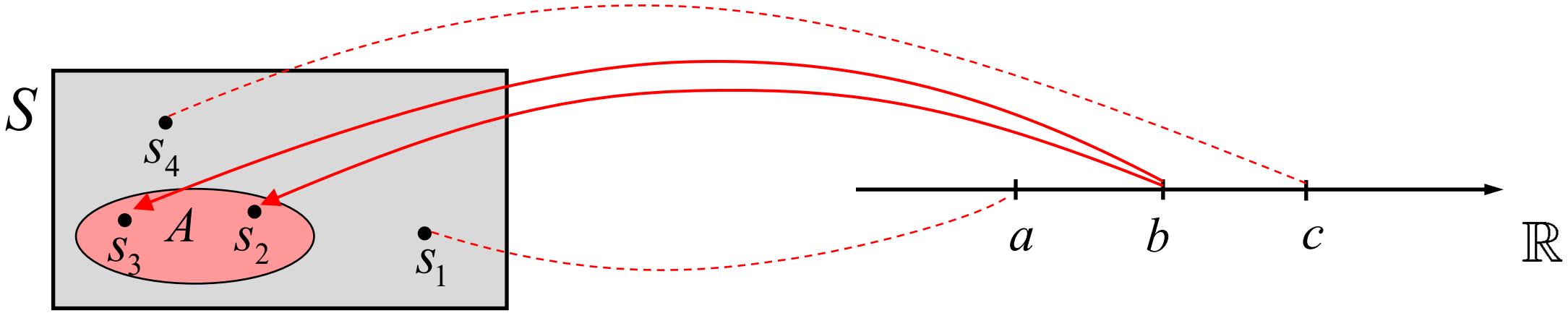
1,1	1,2	1,3	1,4	1,5	1,6
2,1	2,2	2,3	2,4	2,5	2,6
3,1	3,2	3,3	3,4	3,5	3,6
4,1	4,2	4,3	4,4	4,5	4,6
5,1	5,2	5,3	5,4	5,5	5,6
6,1	6,2	6,3	6,4	6,5	6,6



Random Variables

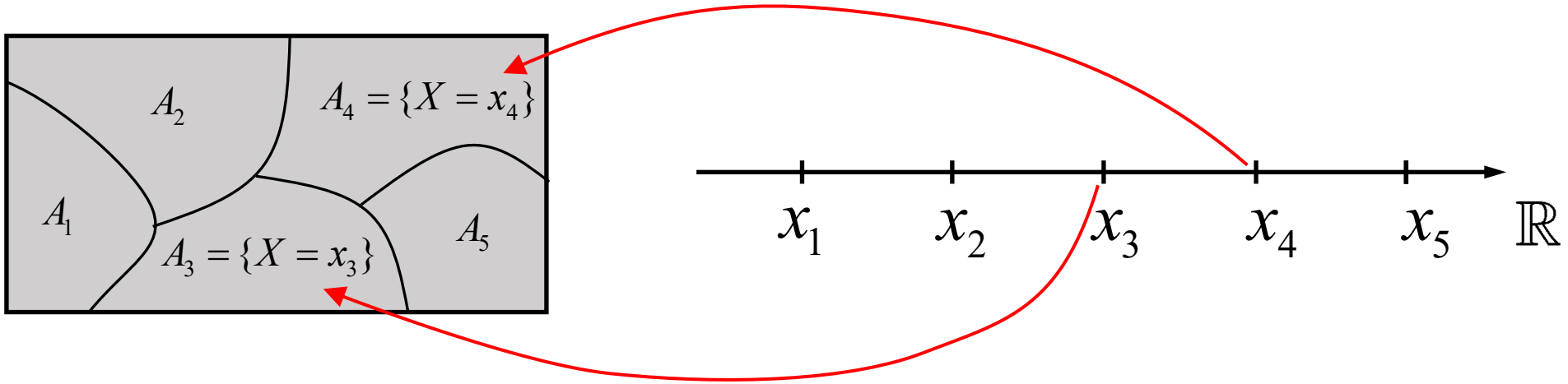
- RVs are used "in reverse" to *describe events*
  - E.g., set of all outcomes for which RV X takes value b

$$A = \{s \in S : X(s) = b\} = \{X = b\}$$



Discrete RVs

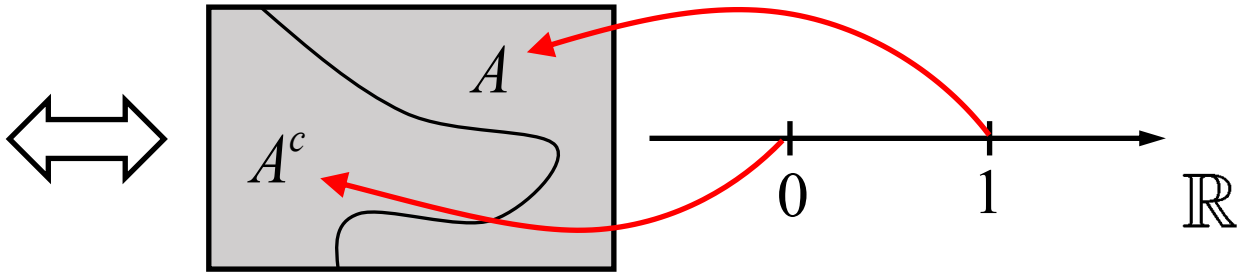
- RV X called *discrete* if it can assume a *finite* $\{x_1,...,x_n\}$ or *countably infinite* $\{x_1,x_2,...\}$ number of values
  - Discrete RVs *partition* sample space in countable # of events
  - E.g., for $X \in \{x_1,\ldots,x_5\}$ define $A_i = \{X = x_i\}, \forall i = 1,\ldots,5$



Indicator RVs

- The *Indicator* RV of an event A, denoted by I_A, takes value 1 when A occurs and 0 otherwise

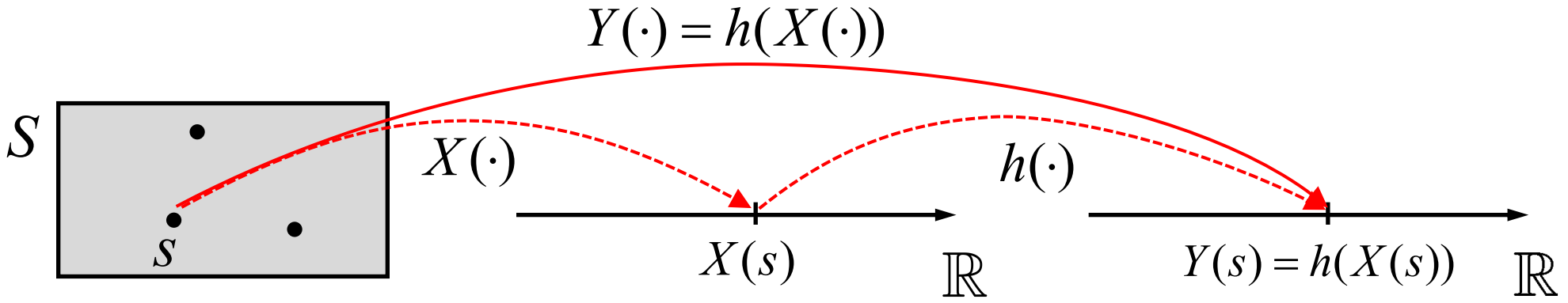
$$I_A(s) = \begin{cases} 1, & s \in A \\ 0, & s \notin A \end{cases}$$



  - Indicator RV partitions sample space in *two events*
  - Indicators connect events to RVs ($A \leftrightarrow I_A$)

Functions of RVs

- Functions of RVs *are* *RVs themselves*!
  - E.g. $Y(\cdot) = h(X(\cdot))$ for RV $X(\cdot)$ and real function $h(\cdot)$



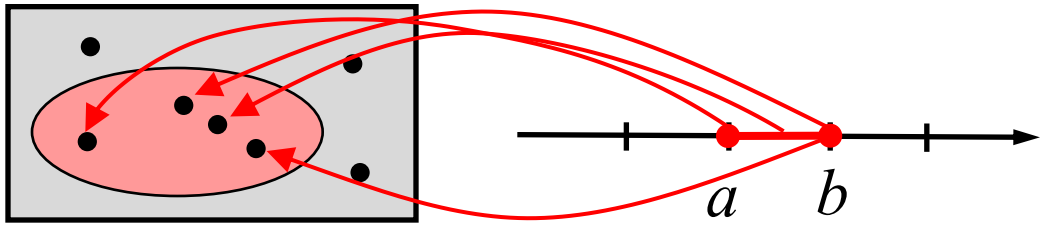
- Similarly for *multivariate* functions
  - E.g. if X, Y are RVs, then $Z = h(X, Y) = X+Y$ is also a RV

Example

- Consider events A, B and their indicator RV's I_A, I_B
  - Show $1-I_A$ is an indicator RV & find its characteristic event
  - Show $I_A \times I_B$ is an indicator RV & find its characteristic event
  - Express the indicator $I_{A \cap B^c}$ as a function of indicator RVs I_A, I_B

RV Distribution

- Events can also be defined by *ranges* of values of RVs
  - E.g., $\{X \in [a,b]\}$ or $\{X \geq 4\}$



- The *distribution* of RV X is the collection of probabilities $P(X \in B)$ for all subsets B of the real line
  - Distribution determines probability of all events defined by X

Discrete Distributions

- Distribution of *discrete* RV $X \in \{x_1, x_2, \ldots\}$ is uniquely determined by probabilities of its individual values

$$p_X(x_i) \equiv P(X = x_i) = P(\{s \in S : X(s) = x_i\}), \quad \forall i = 1, 2, \ldots$$

  - $p_X(\cdot)$ viewed as (discrete) real function, called the *Probability Mass Function* (*PMF*) of RV X

- PMF must satisfy the axioms of probability:
  - $0 \leq p_X(x_i) \leq 1, \quad \forall i$
  - $\sum_{\forall i} p_X(x_i) = 1$

Example

1,1	1,2	1,3	1,4	1,5	1,6
2,1	2,2	2,3	2,4	2,5	2,6
3,1	3,2	3,3	3,4	3,5	3,6
4,1	4,2	4,3	4,4	4,5	4,6
5,1	5,2	5,3	5,4	5,5	5,6
6,1	6,2	6,3	6,4	6,5	6,6

- Roll 2 fair dice & define RV X to be their *sum*; find the PMF of X

Cumulative Distribution Function

- Distribution of *any* RV X can also be determined by its *Cumulative Distribution Function* (*CDF*), defined as

$$F_X(x) \equiv P(X \leq x) = P(\{s \in S : X(s) \leq x\}), \quad \forall x \in \mathbb{R}$$

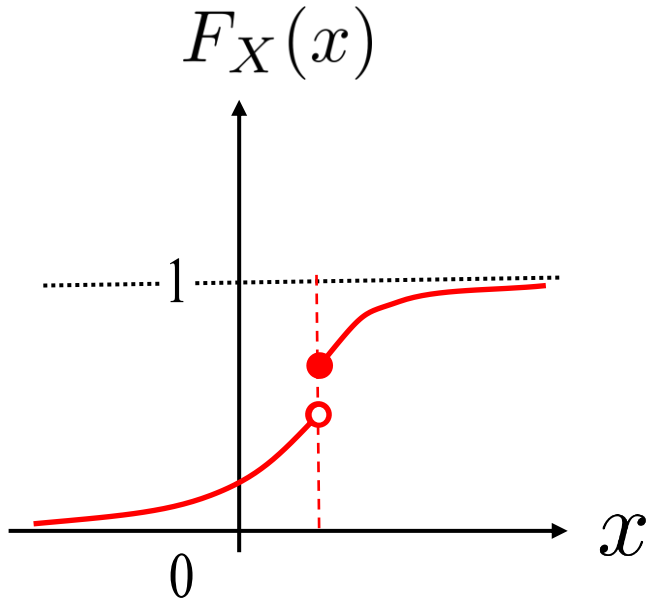
  - Represents probability of RV X being *less than or equal* to value x

- Use CDF to find probability of RV X being in *any* interval $(a,b]$

$$P(a < X \leq b) = P(X \leq b) - P(X \leq a) = F_X(b) - F_X(a)$$

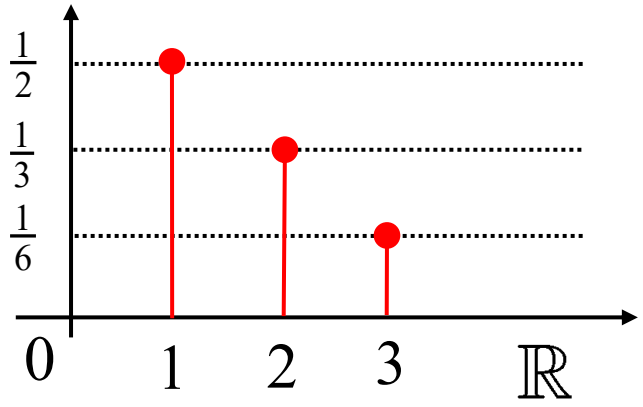
Cumulative Distribution Function

- *Every* CDF must satisfy properties:
  - $F_X(\infty) \equiv \lim_{x \to \infty} F_X(x) = 1$
  - $F_X(-\infty) \equiv \lim_{x \to -\infty} F_X(x) = 0$
  - $\forall x_1 < x_2 \in \mathbb{R} \Rightarrow F_X(x_1) \leq F_X(x_2)$
  - $F_X(\cdot)$ is *right-continuous*
- All derived from probability axioms

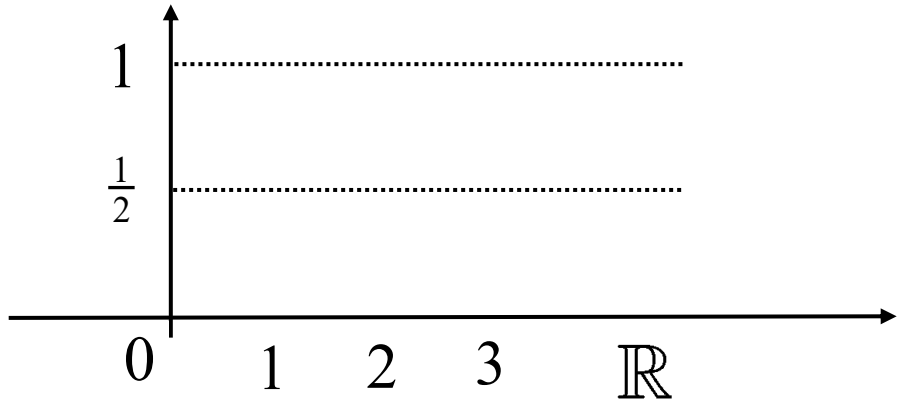


Example

- Consider RV X w/ PMF $p_X(x) = \begin{cases} \frac{1}{2}, & x = 1 \\ \frac{1}{3}, & x = 2 \\ \frac{1}{6}, & x = 3 \end{cases}$ $\Leftrightarrow$



Find and plot CDF of X



Discrete Distributions

- Certain types of RVs arise frequently in problems
  - Consider templates for common applications
  - Calculate probabilities with off-the-shelf PMFs

- Look at common discrete distributions
  - *Bernoulli*
  - *Binomial*
  - *Hypergeometric*

  (also coming up: *Geometric, Negative Binomial, Poisson*)

Bernoulli RV

- Experiments often have binary results
  - E.g., Yes/No, Success/Failure, Heads/Tails etc
- Encode such results with binary RV X, called *Bernoulli* RV

$$X = \begin{cases} 1, & \text{Yes/Success/Heads} \\ 0, & \text{No/Failure/Tails} \end{cases}$$

- Distribution defined by single probability $0<p<1$

$$\begin{cases} p_X(1) = P(X=1) = p \\ p_X(0) = P(X=0) = 1-p \end{cases}$$

Bernoulli Distribution

- Denote that RV X follows Bernoulli distribution as

$$X \sim \text{Bernoulli}(p)$$

  - Tilde symbol (~) stands for "*distributed as*"

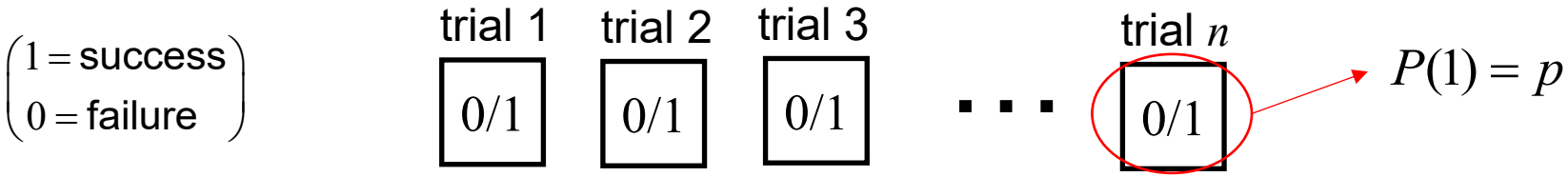
- Bernoulli RVs are equivalent to *indicator* RVs

$$X \sim \text{Bernoulli}(p) \Rightarrow I_{\{X=1\}} \sim \text{Bernoulli}(p)$$

- Bernoulli RVs usually appear in sequences of experiments & serve as building blocks for more complex RVs

Binomial Distribution

- Consider sequence of n *independent binary experiments* (a.k.a. *Bernoulli trials*) with the *same probability* of "success" p



- Let the RV X *count # of successes* in n such trials
  - E.g. (n=5)

    outcome $s = ($ 0 1 0 1 1 $) \Rightarrow X(s) = 3$

- X called *Binomial* RV

Binomial Distribution

- For arbitrary $n \in \mathbb{N}$, what are possible values of Binomial RV X ?

- What is the probability of having $x=3$ successes in $n=5$ trials?

$$\{X = 3\} = \left\{ \begin{array}{ccccc} 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 1 & 1 & 1 \end{array} \right.$$

Binomial Coefficient Properties

- $\binom{n}{0} = \binom{n}{n} = 1$ and $\binom{n}{1} = \binom{n}{n-1} = n$
- $\binom{n}{k} = \binom{n}{n-k}$
- $\binom{n+1}{k+1} = \binom{n}{k} + \binom{n}{k+1}$

Binomial Theorem

- Binomial theorem: $(x+y)^n = \overbrace{(x+y)\cdot\ldots\cdot(x+y)}^{n \text{ times}} = \sum_{i=0}^{n} \binom{n}{i} \cdot x^i \cdot y^{n-i}$

- In particular $\sum_{r=0}^{n} \binom{n}{r} = 2^n$

Binomial Distribution

- More generally, PMF of Binomial RV is

$$p_X(x) = P(X = x) = \binom{n}{x} p^x q^{n-x}, \ \forall x \in \{0,1,2,\ldots,n\} \qquad \begin{pmatrix} \text{where} \\ q = 1 - p \end{pmatrix}$$

  - Denoted $X \sim \text{Binomial}(n,p)$
- Verify this is a valid PMF

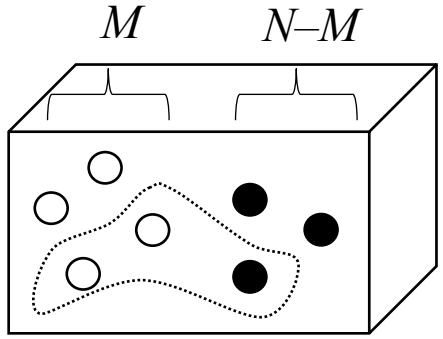
Binomial theorem

$$(x+y)^n = \sum_{i=0}^{n} \binom{n}{i} x^i y^{n-i}$$

Example

- Test has 6 multiple choice questions w/ 4 choices each (A,B,C,D). Student who has not studied answers at random.
  - What is the probability that she get perfect score (6/6)?
  - What is the probability that she passes the test (at least 3/6)?

Hypergeometric Distribution



- Draw #n balls from box with #M white & #N–M black balls, and let RV X be the #*white* balls out of n
  - Find $\mathrm{P}(X=2)$ for $N=7$, $M=4$, $n=3$

- More generally, PMF of *Hypergeometric*(N,M,n) distribution is

$$P(X=x)=\frac{\binom{M}{x}\binom{N-M}{n-x}}{\binom{N}{n}},\ \forall\begin{cases}x\geq\max\{0,n-(N-M)\}\\ x\leq\min\{n,M\}\end{cases}$$